

torch.func.hessian#

<https://pytorch.org/docs/stable/generated/torch.func.hessian.html>

Description:

Computes the Hessian of `func` with respect to the `arg(s)` at index `argnum` via a forward-over-reverse strategy. The forward-over-reverse strategy (composing `jacfwd(jacrev(func))`) is a good default for good performance. It is possible to compute Hessians through other compositions of `jacfwd()` and `jacrev()` like `jacfwd(jacfwd(func))` or `jacrev(jacrev(func))`.

`func` (function) – A Python function that takes one or more arguments, one of which must be a Tensor, and returns one or more Tensors

`argnums` (int or Tuple[int]) – Optional, integer or tuple of integers, saying which arguments to get the Hessian with respect to. Default: 0. Returns a function that takes in the same inputs as `func` and returns the Hessian of `func` with respect to the `arg(s)` at `argnums`.

Function Signature

```
torch.func.hessian(func, argnums=0)[source]#
```

Parameters

Returns

Returns:

Returns a function that takes in the same inputs as func and returns the Hessian of func with respect to the arg(s) at argnums.

Code Examples

```
>>> from torch.func import hessian
>>> def f(x):
>>>     return x.sin().sum()
>>>
>>> x = torch.randn(5)
>>> hess = hessian(f)(x) # equivalent to jacfwd(jacrev(f))(x)
>>> assert torch.allclose(hess, torch.diag(-x.sin()))
```

Notes & Warnings

NOTE

Note You may see this API error out with “forward-mode AD not implemented for operator X”. If so, please file a bug report and we will prioritize it. An alternative is to use `jacrev(jacrev(func))`, which has better operator coverage.

Detailed Documentation

Documentation

Computes the Hessian of `func` with respect to the `arg(s)` at index `argnum` via a forward-over-reverse strategy.

The forward-over-reverse strategy (composing `jacfwd(jacrev(func))`) is a good default for good performance. It is possible to compute Hessians through other compositions of `jacfwd()` and `jacrev()` like `jacfwd(jacfwd(func))` or `jacrev(jacrev(func))`.

`func` (function) – A Python function that takes one or more arguments, one of which must be a Tensor, and returns one or more Tensors

`argnums` (int or Tuple[int]) – Optional, integer or tuple of integers, saying which arguments to get the Hessian with respect to. Default: 0.

Returns a function that takes in the same inputs as `func` and returns the Hessian of `func` with respect to the `arg(s)` at `argnums`.

You may see this API error out with “forward-mode AD not implemented for operator X”. If so, please file a bug report and we will prioritize it. An alternative is to use `jacrev(jacrev(func))`, which has better operator coverage.

A basic usage with a $R^N \rightarrow R^1$ function gives a $N \times N$ Hessian: